

Learning-Based Hybrid Controller for an Underactuated Inverted Pendulum

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Link to project Github repository:
<https://github.com/gkelley3/ecgr-4105-IML>

I. INTRODUCTION

A. Inverted Pendulum

The inverted pendulum is a classical benchmark problem in control theory because it captures many of the challenges associated with nonlinear dynamical systems while remaining mathematically tractable. Although the system is based on the intuitive simple pendulum model, the pendulum dynamics are inherently nonlinear, particularly during large-angle motion. As a result, linear control methods are generally only effective in the local neighborhood of an equilibrium point after linearization.

For the pendulum system considered in this work (Fig. 1), the stable downward equilibrium occurs at $\theta = 0$, while the unstable upright equilibrium occurs at $\theta = \pi$. The primary control objective is to swing the pendulum from any initial condition to the upright position and maintain stability.

An additional challenge arises from limited actuator capabilities in practical pendulum systems. Because real actuators exhibit bounded torque, the controller is generally unable to drive the pendulum directly to the upright equilibrium using purely linear stabilization methods. Instead, the system must exploit the natural nonlinear dynamics of the pendulum to gradually build mechanical energy through repeated oscillations. This underactuated behavior motivates the use of an energy shaping swing-up controller, which regulates the total system energy to guide the pendulum toward the upright equilibrium before a

local stabilizing controller can take over. The energy shaping controller will be discussed further in Section II-B1.

B. Project Objective

To address this challenge, this project implements a hybrid nonlinear control strategy consisting of an energy shaping swing-up controller and a locally stabilizing linear quadratic regulator (LQR). Machine learning techniques are used to predict effective controller gains directly from system conditions (physical parameters and initial conditions), reducing the need for manual tuning and exhaustive parameter searches.

II. METHODOLOGY

A. System Dynamics

The nonlinear pendulum dynamics are given in (1), and the associated system parameters are defined in Table I.

$$\ddot{\theta} = \frac{1}{I} (u - b\dot{\theta} - mgL \sin \theta) \quad (1)$$

TABLE I: Pendulum System Variables and Parameters

Symbol	Description
θ	Pendulum angle
$\dot{\theta}$	Pendulum angular velocity
m	Mass at the end of the rigid massless rod
g	Acceleration due to gravity
L	Length of the rigid massless rod
b	Viscous damping coefficient
I	Pendulum moment of inertia
u	Control input

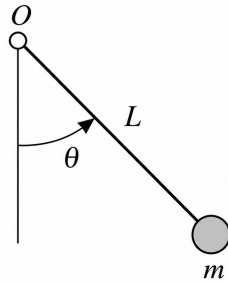


Fig. 1: Pendulum System

Let the system state vector be represented by $x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$. The two equilibrium positions are then defined as:

$$x_{\text{eq, down}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_{\text{eq, up}} = \begin{bmatrix} \pi \\ 0 \end{bmatrix}$$

B. Hybrid Controller

1) *Energy Shaping*: The total mechanical energy in the pendulum system at any time is:

$$E = T + U \quad (2)$$

$$= \frac{1}{2} I \dot{\theta}^2 + mgL(1 - \cos \theta) \quad (3)$$

At the desired final position, $x_{\text{eq, up}}$, (3) reduces to:

$$E_d = 2mgL \quad (4)$$

Let this energy value be denoted as E_d , the desired energy at the final system state. The energy shaping control law used in this work is defined as:

$$u = -k_E \dot{\theta}(E - E_d) \quad (5)$$

where k_E is a proportional gain term that is applied to the error between the current and desired system energy values.

2) *LQR*: While the energy shaping controller is effective for swing-up control, it does not provide stabilization about the upright equilibrium. To stabilize the pendulum near $x_{\text{eq, up}}$, a LQR was implemented using a linearized model of the system dynamics.

The nonlinear pendulum dynamics in (1) were linearized about the upright equilibrium point:

$$x_{\text{eq, up}} = \begin{bmatrix} \pi \\ 0 \end{bmatrix}$$

Let the state error be defined as:

$$\tilde{x} = x - x_{\text{eq, up}}$$

Using the small-angle approximation about $\theta = \pi$,

$$\sin(\theta) \approx -(\theta - \pi)$$

the linearized state-space representation becomes:

$$\dot{\tilde{x}} = A\tilde{x} + Bu \quad (6)$$

where

$$A = \begin{bmatrix} 0 & 1 \\ \frac{mgL}{I} & -\frac{b}{I} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{I} \end{bmatrix}$$

The optimal LQR control law is given by:

$$u = -K\tilde{x} \quad (7)$$

where

$$K = [k_1 \quad k_2]$$

is the optimal feedback gain matrix computed by minimizing the quadratic cost function

$$J_{LQR} = \int_0^\infty (\tilde{x}^T Q \tilde{x} + u^T R u) dt \quad (8)$$

where Q penalizes state deviation and R penalizes control effort.

3) *Hybrid Switching Strategy*: The energy shaping controller is used when the pendulum state is far from the upright equilibrium, allowing the system to gradually accumulate sufficient energy for swing-up. Once the pendulum enters a neighborhood about the upright equilibrium, control authority was switched to the LQR controller for local stabilization.

The switching condition was defined using threshold values on pendulum angle and angular velocity:

$$|\theta - \pi| < \epsilon_\theta, \quad |\dot{\theta}| < \epsilon_{\dot{\theta}}$$

where ϵ_θ and $\epsilon_{\dot{\theta}}$ are user-defined switching tolerances ($\epsilon_\theta = 20^\circ$ and $\epsilon_{\dot{\theta}} = 1.5 \frac{\text{rad}}{\text{s}}$ were chosen for simulations in this project).

C. Machine Learning Framework

The objective of the machine learning component in this work was to reduce the need for manual controller tuning by learning a mapping between pendulum system conditions and effective controller gains. Rather than directly learning the control input u , the machine learning model was trained to predict the hybrid controller parameters.

A 5,000-sample dataset was generated using the Drake Robotics Simulator for each simulated pendulum configuration. A grid search was performed over the controller parameters:

$$k_E, \quad K = [k_1 \quad k_2]$$

where k_E is the energy shaping gain and k_1, k_2 are the LQR feedback gains.

Each controller candidate was evaluated through closed-loop simulation, and several performance metrics were recorded for each trial, including:

- swing-up success or failure
- settling time, T_s
- upright stabilization error, e_{up}
- control effort, U

These recorded quantities were then combined into a weighted performance cost function of the form:

$$J = w_1 T_s + w_2 e_{\text{up}} + w_3 U \quad (9)$$

where w_1, w_2 , and w_3 are weighting coefficients. The controller gains producing the minimum cost were selected as the optimal labels for the dataset. In this work, the upright error weight w_2 was set much larger than the other weighting coefficients due to the desired control objective. It may be necessary to adjust these weights appropriately in other applications.

The input feature vector for the machine learning model consisted of the pendulum system parameters and initial conditions:

$$x_{\text{ML}} = [m \quad L \quad b \quad u_{\text{max}} \quad \sin \theta_0 \quad \cos \theta_0 \quad \dot{\theta}_0]$$

where $u_{\max}$ represents the actuator limits. $\sin \theta_0$ and $\cos \theta_0$ were included as input features to uniquely encode the states corresponding to similar angles, such as $\theta = \pi$ and $\theta = -\pi$, for example. The output labels were the corresponding optimal controller gains:

$$y_{\text{ML}} = [k_E \quad k_1 \quad k_2]$$

Initially, a multilayer perceptron (MLP) regressor was trained to jointly predict all controller gains. While the model produced acceptable predictions for the energy shaping gain k_E , the resulting coefficient of determination (R^2) values for the LQR gains k_1 and k_2 indicated weak predictive performance. This behavior suggested that the nonlinear relationship between the system parameters and optimal LQR gains was not adequately captured by the initial neural network architecture.

To improve prediction accuracy, a random forest regressor was implemented for the LQR gain prediction task. The ensemble-based structure of the random forest provided improved performance for the more complex gain relationships while maintaining robust generalization across the dataset. The final machine learning framework therefore used separate models for different controller components: an MLP regressor for predicting k_E and a random forest regressor for predicting the LQR gains k_1 and k_2 .

The dataset was divided into training and testing subsets, and model performance was evaluated using regression metrics including mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2). The predicted gains generated by the trained models were then applied directly within the hybrid controller framework and compared against the original grid-search-generated controller gains through closed-loop simulation.

III. RESULTS AND ANALYSIS

A. Dataset

A 5,000-sample dataset was generated through closed-loop simulation in Drake using randomized pendulum parameters and controller gain grid searches. For each system configuration, the controller gains minimizing the weighted performance cost function were selected as the dataset labels.

Fig. 2 shows the correlation matrix for the dataset variables. Moderate correlations were observed between the physical pendulum parameters and the optimal controller gains, indicating that the mapping between system properties and controller tuning parameters is nonlinear and coupled. This behavior motivated the use of machine learning models capable of approximating nonlinear relationships.

Figs. 3–5 show the distributions of the optimal controller gains identified through the grid-search procedure. The gain distributions were relatively broad, demonstrating that different pendulum parameter combinations required substantially different controller tuning values. Additionally, the distributions were not perfectly Gaussian, particularly for the LQR gains, suggesting that simple linear regression approaches would likely be insufficient for accurate gain prediction.

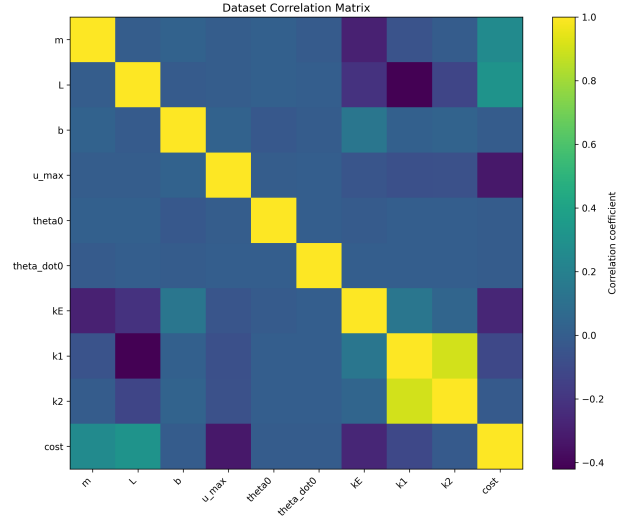


Fig. 2: Dataset Correlation Matrix

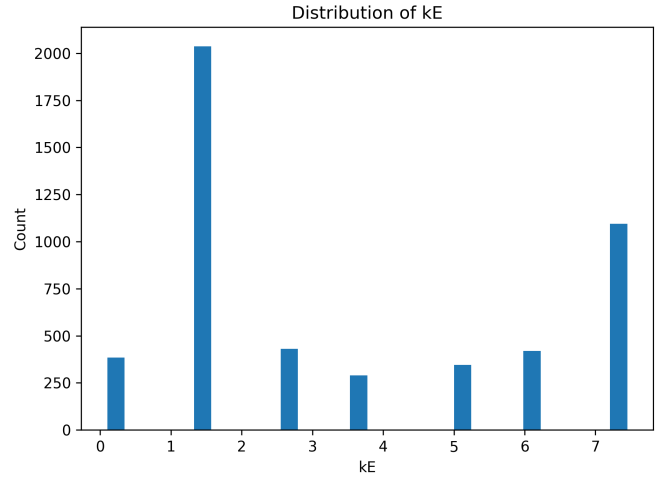


Fig. 3: Dataset Distribution of k_E Values

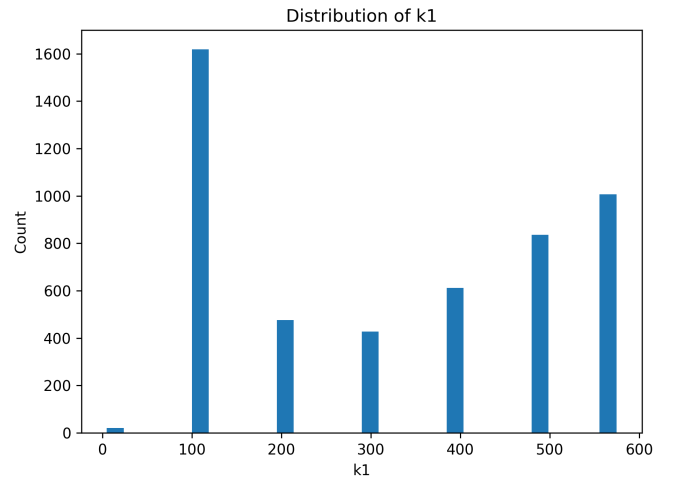


Fig. 4: Dataset Distribution of k_1 Values

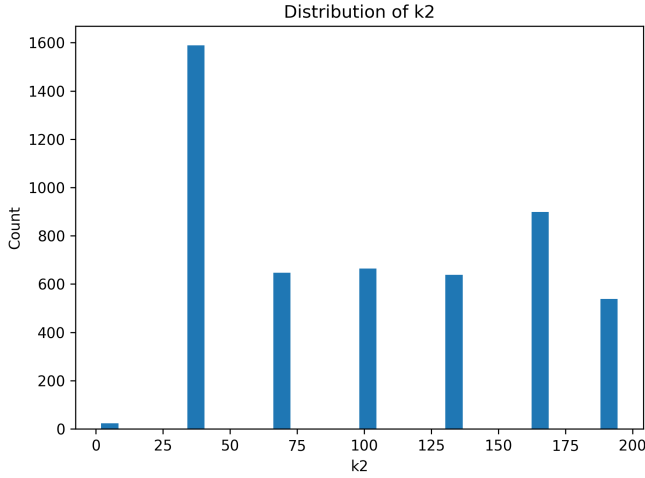


Fig. 5: Dataset Distribution of k_2 Values

B. Model Training

The machine learning models were evaluated using mean absolute error (MAE), root mean squared error (RMSE), normalized MAE, and the coefficient of determination (R^2). The resulting prediction performance metrics are summarized in Table II. The trained models achieved moderate predictive performance across all controller gains. The energy shaping gain k_E produced an R^2 value of 0.4383, while the LQR gains k_1 and k_2 achieved R^2 values of 0.4886 and 0.3923, respectively. Although these values indicate that the learned mapping was only partially captured, the normalized MAE values remained below approximately 20% for all gains, suggesting that the predicted gains remained reasonably close to the optimal grid-search labels.

TABLE II: Gain Prediction Performance Metrics

Gain	MAE	RMSE	R^2	Normalized MAE
k_E	1.5016	2.0026	0.4383	0.2043
k_1	100.4982	132.5771	0.4886	0.1765
k_2	33.7798	45.3918	0.3923	0.1756

As outlined in Section II-C, during development, a single MLP regressor was initially trained to jointly predict all controller gains. However, the resulting prediction accuracy for the LQR gains was relatively weak. To improve performance, a random forest regressor was implemented for predicting k_1 and k_2 , while the MLP was retained for predicting the energy shaping gain k_E . This hybrid machine learning approach produced more consistent predictive behavior across the full controller parameter set.

C. Closed-Loop Controller Performance

The predicted controller gains were evaluated through closed-loop simulation and compared against the original dataset grid-search gains. The resulting performance metrics are summarized in Table III.

TABLE III: Closed-Loop Performance Comparison

Metric	Dataset Grid-Search Gains	ML-Predicted Gains
Success Rate	1.0000	1.0000
Mean Cost	68.7195	70.9771
Mean Settling [s]	2.0262	2.1186
Mean Error	6.3778	6.5903
Mean Effort	29.1525	29.5579

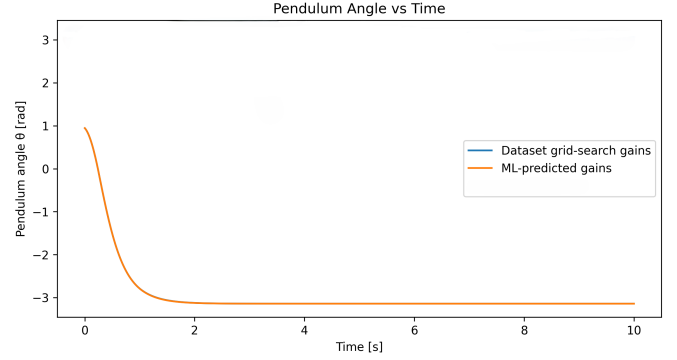


Fig. 6: Pendulum Angle θ vs. Simulation Time – Representative Sample

Both the original grid-search gains and the machine-learning-predicted gains achieved a 100% swing-up success rate across the evaluation trials. Although the ML-predicted gains produced slightly higher mean cost, settling time, stabilization error, and control effort, the overall closed-loop behavior remained very similar to that achieved using the optimal grid-search gains.

The closed-loop response plots demonstrate that the ML-predicted controller gains produced behavior nearly identical to the dataset grid-search gains. In the pendulum angle trajectories, both controllers successfully achieved swing-up and stabilized the pendulum near the upright equilibrium with comparable settling times and minimal steady-state error. Similarly, the control input plots show that the ML-predicted controller generated torque commands with magnitude and

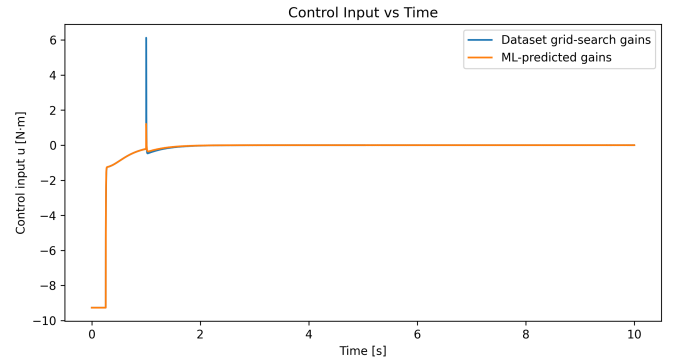


Fig. 7: Control Input vs. Simulation Time – Representative Sample

overall structure similar to the grid-search controller, indicating that the learned model preserved the essential control policy despite moderate gain prediction error. These results suggest that exact recovery of the original gain labels was not necessary to achieve high-quality closed-loop performance, as multiple neighboring gain combinations produced similar system behavior.

These results indicate that exact gain prediction was not the most meaningful evaluation metric for this application. The grid-search labels were generated from a coarse discrete search space, and many neighboring gain combinations produced nearly identical closed-loop behavior. Consequently, numerically different controller gains could still produce successful stabilization and comparable transient response characteristics.

The relatively small degradation in closed-loop performance despite moderate regression accuracy demonstrates that the machine learning framework was able to capture the underlying controller tuning trends sufficiently well for practical stabilization performance. This result suggests that machine learning may provide a viable approach for reducing manual controller tuning effort in nonlinear hybrid control systems.

IV. CONCLUSION

This work presents a hybrid nonlinear control framework for the inverted pendulum, consisting of an energy shaping swing-up controller and a locally stabilizing LQR controller. A machine learning framework was then developed to approximate the controller gain-selection process directly from pendulum system parameters.

The trained machine learning models were able to successfully approximate the underlying gain-selection policy generated through exhaustive simulation and grid-search optimization. Although the regression models achieved only moderate R^2 values, the resulting closed-loop system performance remained nearly identical to that achieved using the original grid-search-generated gains. Both controller implementations consistently achieved successful pendulum swing-up and stabilization with only minor differences in settling time, stabilization error, and control effort.

An important observation from this work was that exact gain prediction was not the most meaningful evaluation metric for controller performance. Because the dataset was generated from a coarse discrete grid search, many neighboring controller gain combinations produced very similar closed-loop system behavior. Consequently, numerically different gains were often capable of producing nearly equivalent stabilization performance. This result demonstrated that closed-loop simulation metrics provided a more meaningful measure of controller quality than raw regression error alone.

The machine learning approach additionally provided a significant computational advantage compared to exhaustive gain search methods. Once trained, the models were capable of producing controller gain predictions nearly instantaneously, whereas traditional grid-search-based tuning required repeated closed-loop simulations over large parameter spaces.

Several opportunities exist for future work. One possible extension would involve replacing the coarse discrete grid search with continuous optimization techniques to generate smoother and more accurate controller labels. More advanced control and learning strategies, such as reinforcement learning or model predictive control (MPC), could also be explored for nonlinear swing-up and stabilization tasks. Finally, the framework developed in this work could be extended to higher-dimensional nonlinear robotic systems, including balancing robots and bipedal locomotion platforms, where manual controller tuning becomes increasingly difficult due to system complexity.

REFERENCES

- [1] R. Tedrake, "Underactuated robotics: Algorithms for walking, running, swimming, flying, and manipulation," 2009. MIT OpenCourseWare, Section 3.6: Energy Shaping.